Multi-Modal Semantic Item Identifiers for Generative Recommendation

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Abstract

We study multi-modal semantic item identifiers for generative recommendation. Holding the generative retriever fixed, we perform a *lightweight* multi-modal intervention by injecting CLIP-based visual features to isolate the effect of visual semantics. Experiments on Amazon domains show that visual signals improve retrieval in appearance-driven categories but offer limited or inconsistent gains in text-centric ones. We further observe a diversity–fairness trade-off: multi-modal identifiers broaden item and brand coverage, yet they can amplify exposure inequality. Resources are available at https://anonymous.4open.science/r/MMGenRec-24F8.

CCS Concepts

• **Computing methodologies** → **Neural networks**; • **Information systems** → **Recommender systems**; *Information retrieval*.

Keywords

Generative recommendation, Semantic identifiers, Multi-modal fusion, Fairness and diversity

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1 Introduction

Generative recommendation (GenRec) treats item retrieval as sequence generation: given a user history, the model generates a discrete identifier for the next item rather than ranking candidates [17, 18, 21]. A common approach represents items using *semantic identifiers*, which are discrete codes obtained by quantizing continuous item representations, and trains a transformer to predict the next identifier based on user interactions [21].

Most existing work constructs these identifiers using text-derived signals and evaluates performance mainly in terms of accuracy. However, many recommendation domains are inherently multi-modal. Product images often encode information such as color, shape, and style that is not fully captured by text metadata. This raises a central question: *can visual semantics improve semantic identifiers without changing the downstream generative retriever?*

Answering this question is challenging because prior multi-modal GenRec methods modify multiple components at once. For instance, VIP5 [5], UniMP [25], PerVRA [10], and VIRAL [26] introduce additional fusion modules, objectives, or training strategies. While these approaches improve performance, they make it difficult to attribute gains specifically to visual information.

In this work, we focus on a controlled setting where only the construction of semantic identifiers is changed. We combine CLIP-derived image features with text embeddings before quantization, while keeping the generative retriever fixed. This allows us to isolate the contribution of visual semantics without introducing additional inference-time cost. Across multiple Amazon domains, we observe that the impact of visual signals varies across domains. Gains tend to be larger in domains where appearance is important, while improvements are smaller or inconsistent in text-centric domains.

We further examine beyond-accuracy effects. Multi-modal identifiers tend to increase catalog and brand coverage, indicating broader item exposure. At the same time, they can lead to more concentrated exposure over a subset of items, suggesting a trade-off, where broader coverage can come with more concentrated exposure.

Contributions. (i) *Controlled empirical study.* We analyze multi-modal semantic item identifiers under a fixed GenRec retriever, isolating the effect of visual semantics during identifier construction. (ii) *Domain-dependent effects.* We observe that visual features tend to provide larger gains in appearance-driven domains and limited or inconsistent effects in text-centric settings. (iii) *Beyond-accuracy analysis.* We study coverage and exposure inequality, showing that multi-modal identifiers can expand catalog reach while affecting exposure distribution.

2 Related Work

Transformer-based sequential recommenders such as SASRec [9] and S^3-Rec [31] model user interaction histories to rank candidate items. In contrast, generative recommendation formulates retrieval as sequence generation, where models directly predict discrete item identifiers instead of scoring candidate sets [2, 15, 27, 32]. Semantic identifier approaches instantiate this idea by learning quantized codes from item representations and training a transformer to generate these codes from user histories [21, 22]. Our work builds on this line by focusing on how the signals used to construct identifiers affect downstream behavior.

Recent work extends generative recommendation to multi-modal settings by incorporating both text and visual information. These methods typically introduce additional components such as fusion modules, modality-specific encoders, or auxiliary training objectives [11, 28, 33]. While they demonstrate improved effectiveness, changes to both representations and model design make it difficult to isolate the contribution of visual features. In contrast, we study a controlled setting where only identifier construction is modified, allowing us to examine the effect of visual semantics without confounding architectural changes.

Beyond-accuracy evaluation has been widely studied in recommendation systems. Prior work highlights that optimizing for

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accuracy alone can lead to reduced catalog coverage and increased exposure concentration, motivating the use of additional metrics such as item and brand coverage, as well as inequality measures like the Gini coefficient [8, 30]. These findings show that improvements in effectiveness can introduce trade-offs in diversity and fairness, often concentrating exposure on a small subset of items. While such evaluations are common in collaborative and sequential recommenders, they are less explored in generative settings, especially under multi-modal signals. In this work, we extend these analyses to GenRec and study how multi-modal semantic identifiers affect both effectiveness and exposure distribution.

3 Methodology

We formalize GenRec and describe how semantic identifiers are constructed and used for retrieval. To enable a controlled comparison, we fix the retriever (TIGER [21]) and vary only the semantic signals used to construct identifiers. We train separate RQ-VAE codebooks for each setting (text-only, image-only, add, concat) using the same training and decoding procedure.

3.1 Problem formulation

Let $\mathcal{U}$ and $\mathcal{I}$ denote sets of users and items. Each user $u \in \mathcal{U}$ has an interaction history $(i_1, \ldots, i_n)$ with $i_j \in \mathcal{I}$, and the task is to predict the next item i_{n+1}. GenRec models generate a discrete *semantic identifier* (SID) for the next item by minimizing the negative log-likelihood:

$$\min_{\theta} \sum_{u \in \mathcal{U}} -\log P_\theta\big(\mathrm{SID}(i_{n+1}) \mid \mathrm{SID}(i_1), \ldots, \mathrm{SID}(i_n)\big), \tag{1}$$

where θ are model parameters. In our setup, $\mathrm{SID}(i)$ is a tuple of L discrete codes produced by an RQ-VAE, and we set $L = 4$.

3.2 Unimodal semantic identifiers

The unimodal baseline (text-only and image-only) follows TIGER [21] (Figure 1). Item features are encoded into a continuous representation $\tilde{\mathbf{x}} \in \mathbb{R}^d$ using either text or image encoders. Specifically, we use Sentence-T5 [16] for text (e.g., title, brand, category) and CLIP [20] for images. The resulting embedding is quantized using an RQ-VAE into a hierarchical code $\mathrm{SID}(i) = (c_1, \ldots, c_L)$, where deeper levels capture more fine-grained information.

User histories are represented as sequences of semantic identifiers. A transformer encoder–decoder generates the next identifier autoregressively by minimizing:

$$\mathcal{L}_{\text{seq2seq}} = -\sum_{t=1}^{L} \log P(c_{\text{next},t} \mid \text{context}, c_{\text{next},<t}), \tag{2}$$

where $c_{\text{next}} = (c_1, \ldots, c_L)$ and $\text{context} = (\mathrm{SID}(i_1), \ldots, \mathrm{SID}(i_n))$.

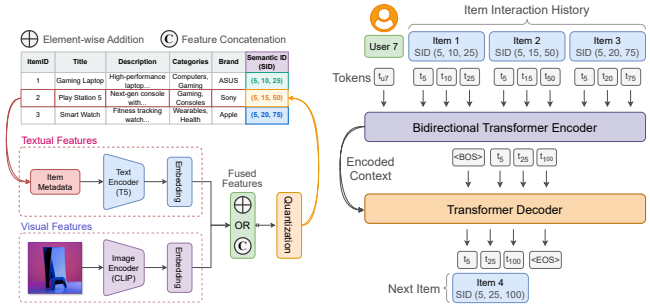
3.3 Multi-modal semantic identifiers

We construct multi-modal semantic identifiers by incorporating visual signals during identifier construction, while keeping the retriever unchanged (Figure 1b).

Each item is encoded using Sentence-T5 for text and CLIP for images:

$$\tilde{\mathbf{x}}_{\text{text}} = \text{Sentence-T5}(s), \tag{3}$$

$$\tilde{\mathbf{x}}_{\text{img}} = \text{CLIP}(v), \tag{4}$$



(a) RQ-VAE semantic IDs. **(b) GenRec retriever [21].**

Figure 1: Overview: (left) semantic identifier construction; (right) seq2seq retriever predicting the next identifier.

where s denotes the item text and v denotes the product image. We ℓ_2-normalize each modality:

$$\mathbf{x}_m = \frac{\tilde{\mathbf{x}}_m}{\|\tilde{\mathbf{x}}_m\|_2}, \quad m \in \{\text{text}, \text{img}\}. \tag{5}$$

We combine the two modalities before quantization using two simple fusion strategies:

$$\mathbf{x}_{\text{fused}} = \begin{cases} \mathbf{x}_{\text{text}} + \mathbf{x}_{\text{img}}, & \text{(add)}, \\ W[\mathbf{x}_{\text{text}}; \mathbf{x}_{\text{img}}], & \text{(concat)}, \end{cases} \tag{6}$$

where $W \in \mathbb{R}^{d \times 2d}$ is a learned projection. Addition assumes that both modalities are already aligned and contribute comparable signals, and can therefore be combined directly. Concatenation allows the model to learn how to weight and combine modality-specific information through the projection layer.

The fused representation is quantized by the RQ-VAE to produce multi-modal semantic identifiers, which are then used by the same retriever as in the unimodal setting. This design ensures that any observed differences in performance can be attributed to the semantic signals used during identifier construction rather than changes in the retriever.

4 Experimental Setup

We evaluate how visual semantics affect generative retrieval across domains with different levels of visual relevance. We compare unimodal and multi-modal semantic identifiers while keeping the retriever fixed, and measure effectiveness, coverage, and exposure inequality.

4.1 Datasets

We use categories from the Amazon Product Review corpus that vary in how much visual information is expected to influence user decisions. From Amazon 2014 [6, 14], we include *Beauty*, *Sports & Outdoors*, and *Toys & Games*, following prior semantic-ID GenRec evaluation [21]. From Amazon 2023 [7], we evaluate *All Beauty*, *Sports & Outdoors*, *Toys & Games*, *Video Games*, and *Software*.

Categories such as *Toys & Games* and *Sports & Outdoors* are more visually driven (e.g., shape or style), while *Software* is treated as text-centric, where functionality is primarily described through text. These categorizations serve as a qualitative guide for analysis.

Following [21], we truncate histories to length 20 and filter users with fewer than five interactions. For Amazon 2023, we sample 25K users per category when available and retain all users for *All Beauty*.

Preprocessing follows the P5 pipeline,[1] with a 95/5 split of item representations for RQ-VAE training/validation and leave-last-out evaluation for the retriever. Dataset statistics are provided in our repository.[2]

4.2 Evaluation metrics

We evaluate performance along multiple dimensions.

Effectiveness. We report Recall@K and NDCG@K for $K \in \{5, 10\}$ [21].

Coverage. We report item and brand coverage at cutoff K (ItemCov@K, BrandCov@K), defined as the fraction of unique items or brands recommended across users in the top-K.

Exposure inequality and user-level disparity. We measure exposure concentration using the Gini coefficient over item (or brand) exposure [1], where lower values indicate more equal exposure. We also report RelDisp@K [3] to capture disparity across users.

4.3 Baselines

We compare against both generative and ranking-based recommendation methods. For GenRec, we use TIGER [21] with unimodal semantic identifiers as the main baseline. For ranking-based methods, we include P5 [4], Caser [24], HGN [13], GRU4Rec [19], BERT4Rec [23], FDSA [29], SASRec [9], and S^3-Rec [31].

4.4 Implementation details

(i) *Codebase:* We build on an open-source RQ-VAE recommender implementation,[3] extend it to Amazon 2023 [7], and apply the leakage fix from [21]. Hyperparameters are tuned on one validation category and then fixed. For multi-modal identifiers, we encode images using CLIP ViT-L/14 [20] and cache embeddings. (ii) *Architecture:* The retriever is a transformer encoder–decoder with hidden size 512 and 8 attention heads. The RQ-VAE uses $L = 4$ residual quantizers, hidden sizes [512, 256, 128], latent size 32, and codebook size 256. (iii) *Training:* The seq2seq retriever is trained for 200K iterations with AdamW [12] (batch size 256, learning rate 3×10^{-4}), with 10K warmup steps and inverse-square-root decay. The RQ-VAE is trained for 400K iterations with AdamW (batch size 64, learning rate 5×10^{-4}).

5 Experimental Results

5.1 Comparison with established baselines

Table 1 compares GenRec with unimodal and multi-modal semantic identifiers against ranking-based sequential recommenders. The goal is to understand when visual semantics improve performance relative to both text-only GenRec and standard baselines.

In *Sports & Outdoors*, product titles such as "yoga mat" or "exercise ball" are often generic and do not clearly distinguish between items. In this setting, text-only identifiers perform substantially worse than sequential baselines, suggesting that different items collapse into similar representations. Adding visual features improves discrimination between such items. This occurs because text-only representations cannot distinguish visually different items with similar descriptions, while visual features introduce additional discriminative signals. The multi-modal (Concat) variant achieves large gains in this category, substantially improving over both text-only GenRec and sequential methods. A similar trend is observed in *Toys & Games*, where visual signals help separate items with similar textual descriptions.

In contrast, in *Beauty*, where textual metadata already captures much of the relevant information, multi-modal identifiers provide smaller and less consistent improvements and do not match the strongest sequential baselines. Multi-modal fusion yields modest NDCG gains but lower Recall@10. Visual information is most useful when it provides signals not captured by text.

5.2 Domain generalization

To assess whether these trends hold beyond Amazon 2014, we evaluate additional categories from Amazon 2023 [7]. Table 2 shows that the effect of visual features varies across domains.

In categories such as *Video Games* and *Toys & Games*, incorporating visual features leads to large improvements over text-only identifiers. These gains are consistent with the idea that images help distinguish items that are otherwise similar in text. In these settings, multi-modal identifiers separate items more effectively, leading to higher retrieval performance.

The relative performance of fusion strategies varies across domains. In some cases, addition performs better, while in others concatenation is more effective. This difference suggests that the usefulness of each strategy depends on whether visual features provide information beyond text or largely overlap with existing textual signals. When the two modalities provide similar information, simple combination is sufficient. When they capture different aspects of an item, a learned projection can better balance their contributions.

Software is used as a text-centric case where images are weakly related to item functionality. In this setting, image-only identifiers perform substantially worse than text-only, and multi-modal fusion only partially recovers performance. Adding visual information can introduce noise when it does not improve item discrimination beyond what is already captured by text.

5.3 Fairness and diversity trade-offs

Table 3 reports beyond-accuracy metrics. Multi-modal identifiers consistently increase catalog reach, but their effect on exposure distribution varies across domains.

The strongest effect is observed in coverage. In Amazon 2023 categories such as *Sports* and *Toys*, text-only identifiers recommend a very small subset of items, leading to near-zero ItemCov. Image-only identifiers increase coverage substantially, and multi-modal identifiers further improve over text-only. This suggests that visual features help expose a broader set of items and brands.

At the same time, increased coverage does not necessarily lead to more balanced exposure. In several cases, multi-modal fusion increases the Gini coefficient, indicating that recommendations become more concentrated on a subset of items. This occurs because visually distinctive items receive more exposure.

User-level disparity and item-level inequality can diverge. In some settings, multi-modal identifiers reduce disparity across users

[1] https://github.com/jeykigung/P5/blob/main/preprocess/data_preprocess_amazon.ipynb
[2] https://anonymous.4open.science/r/MMGenRec-24F8/README.md
[3] https://github.com/EdoardoBotta/RQ-VAE-Recommender

Table 1: Performance comparison on sequential recommendation on the Amazon (2014) dataset. Bold (underline) denotes the best (second-best) result per column across *all* methods. The last row reports the % change (chg.) of the best multi-modal semantic IDs (Add/Concat) over the best *non-multimodal* baseline per column.

Methods	Sports & Outdoors				Beauty				Toys & Games			
	Recall@5	NDCG@5	Recall@10	NDCG@10	Recall@5	NDCG@5	Recall@10	NDCG@10	Recall@5	NDCG@5	Recall@10	NDCG@10
P5 [4]	0.0061	0.0041	0.0095	0.0052	0.0163	0.0107	0.0254	0.0136	0.0070	0.0050	0.0121	0.0066
Caser [24]	0.0116	0.0072	0.0194	0.0097	0.0205	0.0131	0.0347	0.0176	0.0166	0.0107	0.0270	0.0141
HGN [13]	0.0189	0.0120	0.0313	0.0159	0.0325	0.0206	0.0512	0.0266	0.0321	0.0221	0.0497	0.0277
GRU4Rec [19]	0.0129	0.0086	0.0204	0.0110	0.0164	0.0099	0.0283	0.0137	0.0097	0.0059	0.0176	0.0084
BERT4Rec [23]	0.0115	0.0075	0.0191	0.0099	0.0203	0.0124	0.0347	0.0170	0.0116	0.0071	0.0203	0.0099
FDSA [29]	0.0182	0.0122	0.0288	0.0156	0.0267	0.0163	0.0407	0.0208	0.0228	0.0140	0.0381	0.0189
SASRec [9]	0.0233	0.0154	0.0350	0.0192	0.0387	0.0249	0.0605	0.0318	0.0463	0.0306	0.0675	0.0374
S^3-Rec [31]	0.0251	0.0161	0.0385	0.0204	0.0387	0.0244	**0.0647**	0.0327	0.0443	0.0294	0.0700	0.0376
TIGER (text-only)	0.0061	0.0045	0.0091	0.0055	0.0214	0.0157	0.0275	0.0176	0.0227	0.0181	0.0280	0.0198
TIGER (image-only)	0.0087	0.0059	0.0140	0.0076	0.0289	0.0201	0.0409	0.0239	0.0305	0.0215	0.0411	0.0250
Multi-modal semantic IDs (Add)	0.0214	0.0153	0.0290	0.0178	**0.0399**	**0.0283**	0.0563	**0.0336**	0.0287	0.0209	0.0388	0.0242
Multi-modal semantic IDs (Concat)	**0.0431**	**0.0334**	**0.0519**	**0.0363**	0.0287	0.0194	0.0404	0.0232	**0.0604**	**0.0458**	**0.0768**	**0.0511**
% chg. (best MM vs. best non-MM)	+71.71%	+107.45%	+34.81%	+77.94%	+3.10%	+13.65%	-12.98%	+2.75%	+30.45%	+49.67%	+9.71%	+35.90%

Table 2: TIGER (text- and image-only) vs. multi-modal semantic IDs (Add/Concat) on Amazon (2014, 2023). Add/Concat denote fusion strategies used during identifier construction.

Dataset	Category	Text-only		Image-only		Multi-modal (Add)		Multi-modal (Concat)	
		Recall @10	NDCG @10	Recall @10	NDCG @10	Recall @10	NDCG @10	Recall @10	NDCG @10
Amazon (2014)	Beauty	0.0275	0.0176	0.0409	0.0239	**0.0563**	**0.0336**	0.0404	0.0232
	Sports	0.0091	0.0055	0.0140	0.0076	0.0290	0.0178	**0.0519**	**0.0363**
	Toys	0.0280	0.0198	0.0411	0.0250	0.0388	0.0242	**0.0768**	**0.0511**
Amazon (2023)	All Beauty	0.0711	0.0427	0.0593	0.0430	**0.1107**	**0.0723**	0.0593	0.0456
	Sports	0.0891	0.0629	0.0083	0.0049	0.1072	0.0851	**0.1400**	**0.1145**
	Toys	0.0855	0.0651	0.0134	0.0073	**0.2752**	**0.2275**	0.2533	0.1672
	Video Games	0.0150	0.0084	0.0722	0.0496	0.0496	0.0291	**0.1342**	**0.0889**
	Software	**0.3356**	**0.2300**	0.0851	0.0450	0.0920	0.0497	0.1319	0.0771

while increasing concentration over items. Finally, fairness metrics should be interpreted together with coverage. For example, in Amazon 2023 *Sports*, text-only identifiers achieve low disparity but also recommend a very small set of items. In such cases, fairness metrics can be misleading without coverage context.

In *Software*, visual features do not improve coverage or fairness. Multi-modal variants show similar or slightly worse results compared to text-only identifiers. This is consistent with earlier observations that visual features are less useful when they are weakly aligned with the task.

6 Conclusion

We have studied how visual semantics affect generative recommendation by modifying semantic item identifiers while keeping the retriever fixed. This isolates the contribution of visual signals without introducing architectural changes.

Across domains, the impact of visual signals varies with their relevance. In categories where appearance matters, incorporating images improves item discrimination and retrieval performance. In more text-centric settings, the benefits are smaller and can be inconsistent, and visual signals may introduce noise when they are not informative.

Beyond effectiveness, multi-modal identifiers increase catalog and brand coverage, exposing a broader set of items, but can also

Table 3: Fairness and diversity trade-offs (K=10). Lower is better for Gini/RelDisp; higher is better for ItemCov/BrandCov.

Dataset	Category	Method	Gini↓	RelDisp↓	ItemCov↑	BrandCov↑
Amazon (2014)	Beauty	Text-Only	0.1505	0.6403	0.8789	0.9015
		Image-Only	**0.1009**	**0.2624**	0.8471	0.9059
		MM (Add)	0.2006	0.5406	**0.9023**	**0.9399**
		MM (Concat)	0.2029	0.5788	0.8966	0.9296
	Sports	Text-Only	0.1831	0.3422	0.8101	0.8717
		Image-Only	**0.1672**	0.5417	0.7642	0.8532
		MM (Add)	0.2314	**0.1518**	0.8547	0.9032
		MM (Concat)	0.2215	0.2026	**0.8702**	**0.9128**
	Toys	Text-Only	0.1331	**0.0661**	0.8812	0.9155
		Image-Only	**0.1071**	0.1764	0.7479	0.8527
		MM (Add)	0.2265	0.1411	0.9284	**0.9526**
		MM (Concat)	0.2303	0.0745	**0.9330**	0.9476
Amazon (2023)	All Beauty	Text-Only	0.1085	0.8841	0.9347	0.9316
		Image-Only	0.1188	**0.3467**	0.9322	0.9451
		MM (Add)	0.0967	1.0476	0.9270	0.9244
		MM (Concat)	**0.0927**	0.6041	**0.9408**	**0.9536**
	Sports	Text-Only	0.1768	**0.0494**	0.0010	0.0027
		Image-Only	**0.1710**	1.1127	**0.6903**	**0.7122**
		MM (Add)	0.1716	0.1958	0.1248	0.2119
		MM (Concat)	0.3103	0.1109	0.1340	0.2222
	Toys	Text-Only	0.3821	0.0170	0.0021	0.0055
		Image-Only	0.3453	0.3851	**0.4600**	**0.5526**
		MM (Add)	**0.3018**	0.0197	0.1958	0.2760
		MM (Concat)	0.3605	**0.0160**	0.1862	0.2777
	Video Games	Text-Only	0.3672	**0.1813**	0.6499	0.7564
		Image-Only	0.3077	1.0691	0.6572	0.7060
		MM (Add)	0.2346	0.4408	0.6537	0.7506
		MM (Concat)	**0.2231**	0.3241	**0.6663**	**0.7753**
	Software	Text-Only	**0.0000**	**0.0690**	**0.6180**	0.5000
		Image-Only	0.0022	0.2935	0.6087	**0.6667**
		MM (Add)	**0.0000**	0.0963	0.5563	**0.6667**
		MM (Concat)	**0.0000**	0.0878	0.5759	**0.6667**

lead to more concentrated exposure, indicating a trade-off between coverage and exposure distribution.

Overall, visual signals are useful when they provide complementary information to text, but their contribution is not uniform across domains. Future work should explore adaptive fusion strategies that account for modality relevance.

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